

CP Analysis Report: US Census Bureau Analysis

Stat 184

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Introduction

The U.S. Census Bureau provides some of the most widely used public data for understanding demographic, economic, and social change in the United States. For this project, we use data accessed through the `{tidycensus}` package in R to examine how selected county-level indicators changed in Maryland around the COVID-19 period, especially from 2019 to 2022. This period is important because it includes major disruptions to work, commuting, income, housing, and household behavior.

Our analysis focuses on Maryland counties using American Community Survey (ACS) 1-year and 5-year estimates, as well as the Decennial Census. We use this information to make tables, plots, and narratives explaining some of the changes that have appeared in the state, and across its counties.

Data Provenance:

In this analysis project, we will be working with the `{tidycensus}` package in R. This package in R is designed to make it easy to access US Census data. Firstly, it connects directly to the US Census Bureau API with an API Key given directly to you by the US Census Bureau. Using that key, it will return data in a format that is compatible with `{tidyverse}`. (i.e data frames that can be wrangled, cleaned, and easily used for data analysis).

Where the data comes from:

All data accessed through `{tidycensus}` comes from the following:

Decennial Census

- Conducted every 10 years (e.g., 2010, 2020)
- Attempts to count every person in the U.S.
- Collects basic info: age, sex, race, household relationships, etc

American Community Survey (ACS)

- A continuous survey
- Provides detailed socioeconomic data of the US

Who collected it?

The US Census Bureau (Department of Commerce)

Why was it collected?

- Allocate federal funding
- Determine political representation
- Support research, planning, and policy decisions

How is it collected?

Decennial Census

- Mail forms, online responses, in-person follow-ups
- Aims for a complete count

American Community Survey (ACS)

- Uses sampling, not a full count
- About 3–3.5 million households per year are surveyed (randomly selected)

Random sampling survey techniques include:

- mail surveys
- phone interviews
- in-person follow-ups

When was it collected?

Decennial Census

- Every 10 years (i.e 2020)

American Community Survey (ACS)

- Conducted continuously every year since 2005
- Data is released as 1-year estimates (large populations) or 5-year estimates (smaller, more reliable populations)

For this analysis, we will be using the American Community Survey (ACS) data from the US Census Bureau to answer the following question(s):

- How is the rise of remote work associated with income, internet access, and rent burden across Maryland counties from 2019-2022?
- Are Maryland counties with larger elderly populations (65+) experiencing higher economic strain from 2019-2022?

- How did Income Inequality change, as measured by the Gini Index, during the periods of 2019-2022?
- How did the number of people working from home change during the COVID-19 Pandemic?
- etc.

FAIR/CARE Principles:

The `{tidycensus}` package and the U.S. Census data it accesses align closely with both the FAIR and CARE principles for responsible data use. In terms of FAIR, the data is highly findable, as each variable is assigned a unique identifier (such as `B19013_001` for median income) and can be easily located using tools like `load_variables()`. It is also accessible, since the Census Bureau provides open access through an API that can be queried directly in R with minimal barriers. The data is inter-operable because it is delivered in consistent, standardized formats that integrate seamlessly with tools in the tidyverse, allowing for efficient data wrangling and analysis. Additionally, it is reusable, as the Census provides thorough documentation and maintains consistent variable definitions across years, enabling reliable longitudinal studies like the 2019–2022 county-level analysis conducted here.

From a CARE perspective, Census data supports collective benefit by being publicly available for research, policy, and social understanding. While individual-level control over data use is limited, the Census Bureau maintains privacy protections, addressing authority to control. Ethical considerations are supported through the use of aggregated, de-identified data, though we must still be mindful when analyzing demographic patterns to avoid harmful generalizations.

Variables:

The *raw* attributes we will be focusing on for this project are:

- County
- Year
- Total population
- Total households
- Total renter-occupied households
- Median household income
- Labor force (population in labor force)
- Total workers (commuting / work status)
- Number of unemployed individuals
- Number of people working from home
- Households with internet access

The *constructed/derived* attributes we will be focusing on for this project are:

- Unemployment rate

- Percentage of workers working from home
- Percentage of households with internet access
- Rent burden rate (share of renters spending 30%+ on rent)
- Percentage of population age 65 and older

Data Analysis:

Exploring the Relationship Between Remote Work, Internet Access, Income, and Aging Across Maryland Counties:

Table 1: Maryland County Socioeconomic Indicators (2019)

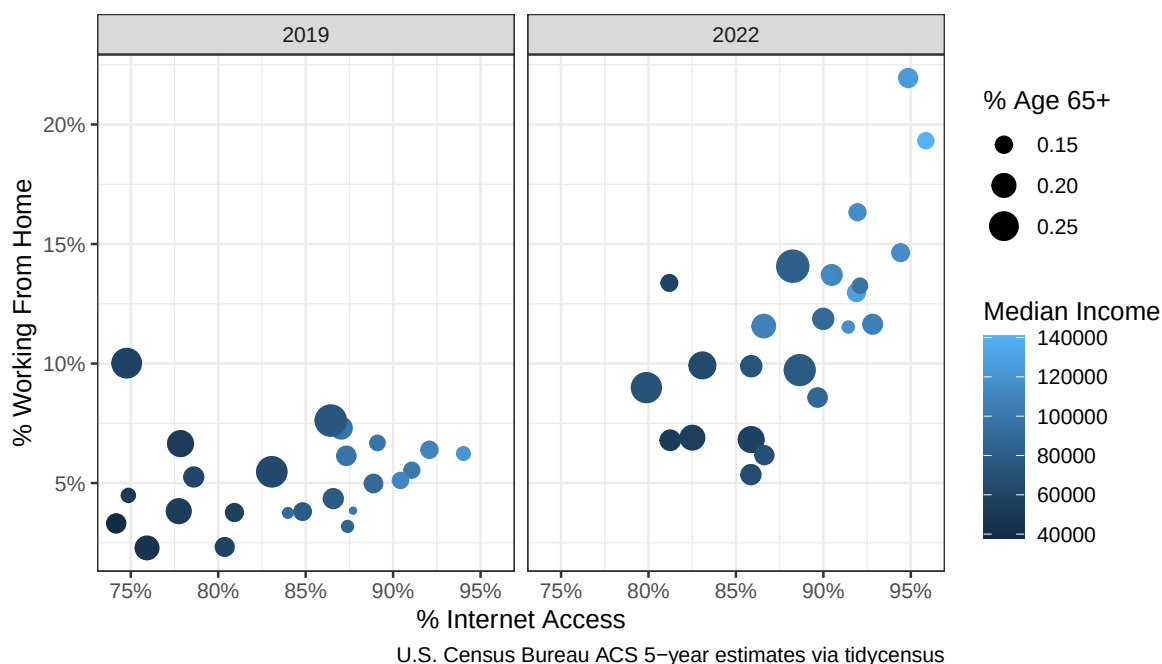
County	Median Income	% WFH	% Internet	Rent Burden	% Age 65+	% Unemp.
Allegany County	\$45,893	2.29%	75.93%	45.88%	19.83%	7.40%
Anne Arundel County	\$100,798	5.54%	91.07%	45.05%	14.40%	4.23%
Baltimore County	\$76,866	4.35%	86.58%	46.94%	16.82%	4.79%
Baltimore city	\$50,379	4.48%	74.86%	48.07%	13.58%	8.29%
Calvert County	\$109,313	5.11%	90.42%	46.26%	14.48%	4.25%
Caroline County	\$58,638	2.33%	80.37%	45.30%	16.03%	4.91%
Carroll County	\$96,769	6.14%	87.33%	44.22%	16.36%	3.34%
Cecil County	\$76,887	3.81%	84.83%	47.46%	15.25%	5.16%
Charles County	\$100,003	3.85%	87.71%	46.64%	12.18%	3.46%
Dorchester County	\$52,917	3.82%	77.74%	48.81%	20.96%	8.02%
Frederick County	\$97,730	6.68%	89.10%	45.46%	14.11%	4.18%
Garrett County	\$52,617	6.65%	77.84%	30.22%	22.01%	3.81%
Harford County	\$89,147	4.98%	88.88%	44.71%	15.80%	4.21%
Howard County	\$121,160	6.24%	94.02%	41.55%	13.41%	3.78%
Kent County	\$58,598	10.02%	74.77%	50.53%	26.21%	4.15%
Montgomery County	\$108,820	6.39%	92.08%	48.65%	15.03%	4.47%
Prince George's County	\$84,920	3.19%	87.40%	49.52%	12.86%	6.01%
Queen Anne's County	\$97,034	7.30%	87.03%	41.72%	18.37%	2.63%
Somerset County	\$37,803	3.31%	74.17%	61.29%	16.22%	8.24%
St. Mary's County	\$89,845	3.75%	83.99%	40.35%	12.55%	3.27%
Talbot County	\$73,547	7.61%	86.44%	45.72%	28.51%	3.36%
Washington County	\$60,860	5.25%	78.60%	40.70%	16.81%	5.47%
Wicomico County	\$56,956	3.77%	80.93%	50.94%	15.45%	7.64%
Worcester County	\$63,499	5.47%	83.06%	46.65%	27.31%	4.91%

Table 2: Maryland County Socioeconomic Indicators (2022)

County	Median Income	% WFH	% Internet	Rent Burden	% Age 65+	% Unemp.
Allegany County	\$55,248	6.90%	82.51%	41.22%	20.79%	6.76%
Anne Arundel County	\$116,009	14.64%	94.42%	45.74%	15.38%	4.16%
Baltimore County	\$88,157	11.87%	89.99%	49.82%	17.62%	5.23%
Baltimore city	\$58,349	13.37%	81.19%	49.15%	14.78%	6.92%
Calvert County	\$128,078	12.97%	91.89%	43.42%	15.50%	3.18%
Caroline County	\$65,326	5.35%	85.86%	48.26%	16.91%	4.61%
Carroll County	\$111,672	13.71%	90.48%	41.99%	17.37%	3.15%
Cecil County	\$86,869	8.58%	89.67%	48.23%	16.37%	4.89%

Charles County	\$116,882	11.53%	91.42%	42.01%	12.90%	4.89%
Dorchester County	\$57,490	6.82%	85.87%	47.41%	21.95%	5.48%
Frederick County	\$115,724	16.33%	91.95%	45.40%	14.90%	3.72%
Garrett County	\$64,447	9.92%	83.08%	26.87%	22.97%	4.63%
Harford County	\$106,417	11.64%	92.82%	43.17%	16.67%	3.69%
Howard County	\$140,971	19.32%	95.86%	43.62%	14.48%	3.78%
Kent County	\$71,635	8.99%	79.88%	49.33%	26.80%	3.28%
Montgomery County	\$125,583	21.95%	94.85%	48.23%	16.15%	4.84%
Prince George's County	\$97,935	13.25%	92.09%	50.33%	14.11%	6.69%
Queen Anne's County	\$108,332	11.57%	86.59%	47.87%	19.68%	3.63%
Somerset County	\$52,149	6.79%	81.24%	50.90%	17.18%	7.04%
St. Mary's County	\$113,668	13.70%	90.28%	38.89%	13.33%	3.65%
Talbot County	\$81,667	14.07%	88.25%	42.73%	29.71%	3.17%
Washington County	\$73,017	9.89%	85.88%	42.95%	17.71%	4.99%
Wicomico County	\$69,421	6.17%	86.63%	47.59%	16.17%	6.87%
Worcester County	\$76,689	9.73%	88.65%	48.02%	28.02%	6.52%

Figure 1: Remote Work, Connectivity, Income, and Age Across Maryland Counties



Long Description: The image is a comparative scatter plot divided into two panels, presenting data from 2019 and 2022 on remote work, internet access, income, and aging across Maryland counties. Each panel displays county data points represented as circles, plotted according to “% Internet Access” on the x-axis and “% Working From Home” on the y-axis. The size of

each circle indicates the percentage of the population aged 65 or older, with larger circles representing higher percentages. Color gradients from dark to light blue depict median income levels, ranging from 40,000 to 140,000. In 2019, data points concentrate at lower working-from-home and internet access percentages, while in 2022, the distribution shifts upwards, indicating an increase in remote work and internet access. We used the `tidycensus` package (Walker, n.d.) and `{ggplot2}` to create this visualization.

(The long description and alt text was produced by ASU-EdPlus-Image-Detail-Tool (“ASU EdPlus,” n.d.) on 5/5/2026)

Figure 1 highlights the relationship between internet access and the proportion of residents working from home across Maryland counties in 2019 and 2022, while also incorporating median income and the share of the population aged 65+. (The data within Figure 1 was created with data from Table 1 and Table 2, look that them for additional information).

From 2019 to 2022, there is a clear upward shift in the percentage of individuals working from home across nearly all counties. In 2019, most counties cluster between roughly 2% and 7% remote work participation. By 2022, this range increases, with many counties falling between 8% and 15%, and some exceeding 20%. This suggests a widespread expansion of remote work, likely influenced by changes during and after the COVID-19 pandemic.

Across both years, there is a positive association between internet access and remote work. Counties with higher levels of internet access tend to exhibit higher rates of working from home. This relationship becomes more pronounced in 2022, where counties with internet access above 90% consistently show elevated remote work participation. This pattern supports the idea that reliable internet infrastructure is a key enabling factor for remote employment.

Income also appears to play a meaningful role. Higher-income counties (indicated by lighter blue shading) are generally located toward the upper-right portion of the plot, reflecting both higher internet access and higher levels of remote work. This suggests that remote work opportunities may be more accessible in higher-income areas, potentially due to differences in job types, education levels, or industry composition.

The size of the points, representing the percentage of the population aged 65 and older, does not show a strong or consistent relationship with remote work. Counties with larger elderly populations are spread throughout the plot, indicating that age composition alone does not strongly determine remote work participation. However, some counties with older populations still exhibit moderate levels of remote work, suggesting that other factors—such as infrastructure and income—may be more influential.

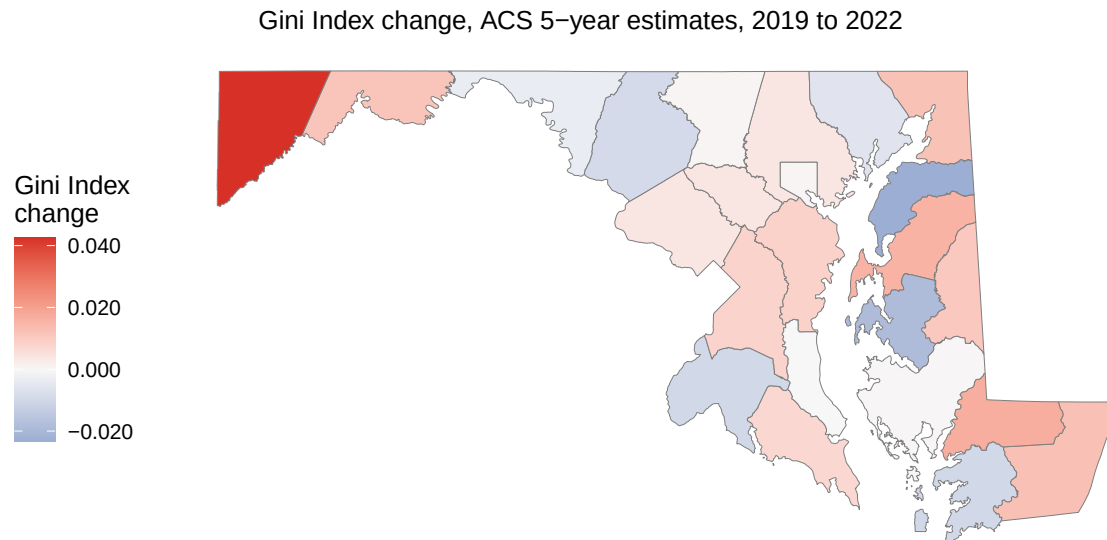
Overall, Figure 1 indicates that increases in remote work from 2019 to 2022 are closely associated with higher internet access and higher median income, while the influence of population age 65+ appears less direct.

Exploring Changes in Income Inequality Across Maryland Counties:

Table 3: Maryland County Rental Housing Stress Indicators, 2022

County	Median Rent	Renters 30%+	Renters 35%+	Renters 50%+	Vacancy
Allegany County	\$743	41.2%	33.4%	17.7%	16.5%
Anne Arundel County	\$1,908	45.7%	36.3%	20.1%	4.9%
Baltimore County	\$1,479	49.8%	41.2%	25.2%	6.0%
Baltimore city	\$1,235	49.1%	41.0%	26.1%	15.8%
Calvert County	\$1,589	43.4%	38.3%	20.9%	6.9%
Caroline County	\$1,074	48.3%	35.9%	19.4%	10.8%
Carroll County	\$1,283	42.0%	35.4%	22.9%	3.8%
Cecil County	\$1,322	48.2%	39.3%	21.7%	10.5%
Charles County	\$1,839	42.0%	36.4%	21.3%	5.1%
Dorchester County	\$968	47.4%	41.5%	22.4%	19.4%
Frederick County	\$1,633	45.4%	36.2%	21.8%	4.2%
Garrett County	\$681	26.9%	19.1%	11.4%	32.7%
Harford County	\$1,475	43.2%	34.7%	21.4%	4.6%
Howard County	\$1,920	43.6%	35.4%	22.4%	3.4%
Kent County	\$1,072	49.3%	37.3%	22.2%	18.8%
Montgomery County	\$1,957	48.2%	39.1%	23.7%	4.5%
Prince George's County	\$1,713	50.3%	40.8%	24.5%	5.0%
Queen Anne's County	\$1,600	47.9%	43.0%	20.5%	9.6%
Somerset County	\$934	50.9%	44.7%	27.9%	23.6%
St. Mary's County	\$1,595	38.9%	34.1%	22.9%	9.8%
Talbot County	\$1,204	42.7%	38.2%	23.2%	17.0%
Washington County	\$1,049	43.0%	33.8%	19.9%	7.3%
Wicomico County	\$1,190	47.6%	39.8%	22.6%	8.6%
Worcester County	\$1,144	48.0%	37.2%	23.9%	58.4%

Figure 2: Change in Income Inequality by Maryland County



Source: U.S. Census Bureau ACS 5-year estimates via tidycensus

Long Description: The image is a choropleth map showing the change in income inequality across Maryland counties between 2019 and 2022. Each county is shaded according to its change in Gini Index value, using a diverging color scale centered at zero. Blue shades represent counties where the Gini Index decreased, indicating a decline in income inequality, while red shades represent counties where the Gini Index increased, indicating a rise in income inequality. Counties shaded closer to white experienced little or no change during the period.

The map includes subtle county boundaries to distinguish individual counties while keeping the focus on the overall geographic pattern. The legend shows the numerical range of Gini Index change. Overall, the visualization shows that changes in income inequality were not uniform across Maryland; some counties experienced increases, while others experienced decreases or minimal change.

(The long description and alt text was produced by ASU-EdPlus-Image-Detail-Tool (“ASU EdPlus,” n.d.) on 5/6/2026)

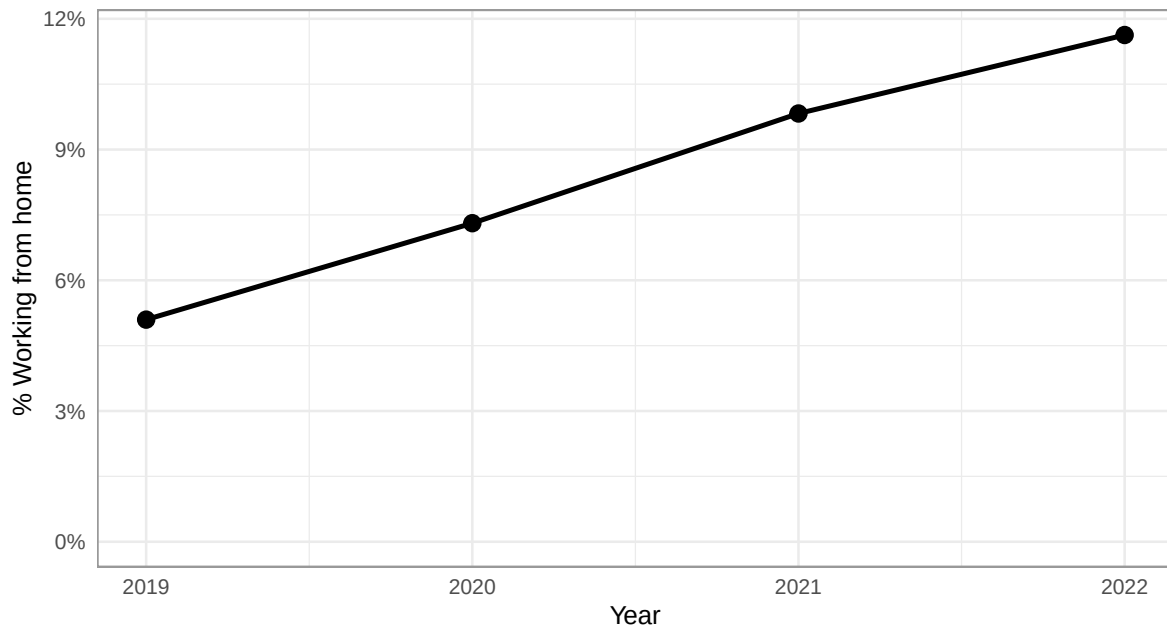
Figure 2 shows that income inequality changed unevenly across Maryland counties from 2019 to 2022. Some counties experienced small increases in the Gini Index, while others experienced small decreases. However, most changes appear modest, suggesting that income inequality remained relatively stable across much of the state.

This goes against our expectation that income inequality would show a significant change after COVID-19. Given the major disruptions to work, income, housing, and household conditions during this period, we expected the Gini Index to shift more noticeably. Instead, the map suggests that any changes in inequality were relatively limited and varied by county.

Overall, Figure 2 suggests that income inequality shifted slightly across Maryland counties, but there was no dramatic statewide increase or decrease.

Examining the Change in Working From Home in Maryland Counties:

Figure 3: Average Work From Home Percentage for Maryland



Source: U.S. Census Bureau ACS 5-year estimates via tidycensus

Long Description: The image is a line chart showing the average percentage of workers working from home across Maryland counties from 2019 to 2022. The x-axis represents year, with points shown for 2019, 2020, 2021, and 2022. The y-axis represents the average percentage of workers who worked from home, ranging from 0% to 12%. Each year is represented by a black circular point, and the points are connected by a black line to show the trend over time.

The chart shows a steady increase in the average work-from-home percentage across Maryland counties during this period. In 2019, the average percentage was slightly above 5%. By 2020, it increased to about 7%, then rose further to nearly 10% in 2021. By 2022, the average reached almost 12%. This upward trend suggests that working from home became increasingly common during and after the COVID-19 pandemic.

(The long description and alt text was produced by ASU-EdPlus-Image-Detail-Tool (“ASU EdPlus,” n.d.) on 5/6/2026)

Figure 3 shows a clear increase in the average percentage of workers working from home across Maryland counties from 2019 to 2022. The average begins slightly above 5% in 2019 and rises steadily, reaching almost 12% by 2022.

This increase likely reflects the impact of the COVID-19 pandemic and the longer-term shift toward remote work. The continued rise through 2022 suggests that working from home became more established rather than simply returning to pre-pandemic levels.

Overall, Figure 3 supports the broader finding that remote work expanded substantially across Maryland during this period.

Author Contribution:

Blake Naluai:

- Software
- Supervision
- Data curation
- Visualization
- Writing: original draft/review & editing
- Formal analysis

Raj Ramnani:

- Software
- Conceptualization
- Data curation
- Visualization
- Writing: original draft/review & editing
- Formal analysis

Code Appendix

```
# Coding Style: tidyverse

# ---- Author: Blake Naluai ----
# ---- Reviewer: Blake Naluai & Raj Ramnani ----

# Load Packages and API key
library(tidycensus)
library(tidyverse)
library(dplyr)
library(tidyr)
library(ggplot2)
library(knitr)
library(kableExtra)
library(scales)
library(esquisse)
library(tigris)
library(sf)

options(tigris_use_cache = TRUE) # permanently store census shapefiles

## Set the Census API key
invisible(census_api_key(
  "6377fd44a462d589653004510dcef417f2f8aabb",
  overwrite = TRUE,
  install = TRUE
))

# ---- Author: Blake Naluai ----
# ---- Reviewer: Blake Naluai & Raj Ramnani ----

## Set variables
vars <- c(
  # Population
  total_population = "B01003_001",
  # Households
  total_households = "B11001_001",
  # Renter-occupied households with income
  renter_occupied_households = "B25070_001",
  # Income
  median_income = "B19013_001",
```

```

# Employment
labor_force = "B23025_003",
total_workers = "B08301_001",
unemployed = "B23025_005",
# Remote work (work from home)
work_from_home = "B08301_021",
# Internet access (household has any internet subscription)
internet_access = "B28002_002",
# Rent burden (30%+ of income, sum these)
rent_30_34 = "B25070_007",
rent_35_39 = "B25070_008",
rent_40_49 = "B25070_009",
rent_50_plus = "B25070_010",
# Age 65+ (we will sum these)
male_65_66 = "B01001_020",
male_67_69 = "B01001_021",
male_70_74 = "B01001_022",
male_75_79 = "B01001_023",
male_80_84 = "B01001_024",
male_85_plus = "B01001_025",
female_65_66 = "B01001_044",
female_67_69 = "B01001_045",
female_70_74 = "B01001_046",
female_75_79 = "B01001_047",
female_80_84 = "B01001_048",
female_85_plus = "B01001_049"
)

## Get data for each year
md_2019 <- get_acs(
  geography = "county",
  state = "MD",
  variables = vars,
  year = 2019,
  output = "wide"
) |>
  mutate(year = 2019)

md_2022 <- get_acs(
  geography = "county",
  state = "MD",
  variables = vars,

```

```

year = 2022,
output = "wide"
) |>
mutate(year = 2022)

## Bind DF's
md_all <- bind_rows(md_2019, md_2022)

## Tidy DF
md_clean <- md_all |>
mutate(
  age_65_plus = male_65_66E + male_67_69E + male_70_74E + male_75_79E +
    male_80_84E + male_85_plusE + female_65_66E + female_67_69E + female_70_74E +
    female_75_79E + female_80_84E + female_85_plusE,
  rent_30_plus = rent_30_34E + rent_35_39E + rent_40_49E + rent_50_plusE
) |>
select(
  County = NAME,
  year,
  total_population = total_populationE,
  total_households = total_householdsE,
  total_renter_occ_households = renter_occupied_householdsE,
  median_income = median_incomeE,
  labor_force = labor_forceE,
  total_workers_commute = total_workersE,
  unemployed = unemployedE,
  work_from_home = work_from_homeE,
  household_internet_access = internet_accessE,
  rent_30_plus,
  age_65_plus
) |>
mutate(
  County = str_remove(County, "\\s*.*$"),
  # Unemployment rate
  unemployment_rate = unemployed / labor_force,
  # % Working from home (proxy)
  pct_wfh = work_from_home / total_workers_commute,
  # Internet access rate
  pct_internet = household_internet_access / total_households,
  # Rent burden (30%+)
  rent_burden = rent_30_plus / total_renter_occ_households,
  # % Age 65+

```



```

    pct_65_plus = age_65_plus / total_population
  ) |>
  arrange(County, year)

# ---- Author: Blake Naluai ----
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md_clean |>
  mutate(
    unemployment_rate = percent(unemployment_rate, accuracy = 0.01),
    pct_wfh = percent(pct_wfh, accuracy = 0.01),
    pct_internet = percent(pct_internet, accuracy = 0.01),
    rent_burden = percent(rent_burden, accuracy = 0.01),
    pct_65_plus = percent(pct_65_plus, accuracy = 0.01),
    median_income = dollar(median_income)
  ) |>
  filter(
    year == 2019
  ) |>
  select(
    "County" = County,
    "Median\nIncome" = median_income,
    "% WFH" = pct_wfh,
    "% Internet" = pct_internet,
    "Rent Burden" = rent_burden,
    "% Age 65+" = pct_65_plus,
    "% Unemp." = unemployment_rate
  ) |>
  kable(
    caption = "Maryland County Socioeconomic Indicators (2019)"
  ) |>
  kable_classic(
    lighttable_options = "striped"
  ) |>
  kable_styling(
    latex_options = "scale_down",
    font_size = 9
  )

# ---- Author: Blake Naluai ----
# ---- Reviewer: Blake Naluai & Raj Ramnani ----

```

```

md_clean |>
  mutate(
    unemployment_rate = percent(unemployment_rate, accuracy = 0.01),
    pct_wfh = percent(pct_wfh, accuracy = 0.01),
    pct_internet = percent(pct_internet, accuracy = 0.01),
    rent_burden = percent(rent_burden, accuracy = 0.01),
    pct_65_plus = percent(pct_65_plus, accuracy = 0.01),
    median_income = dollar(median_income)
  ) |>
  filter(
    year == 2022
  ) |>
  select(
    "County" = County,
    "Median\nIncome" = median_income,
    "% WFH" = pct_wfh,
    "% Internet" = pct_internet,
    "Rent Burden" = rent_burden,
    "% Age 65+" = pct_65_plus,
    "% Unemp." = unemployment_rate
  ) |>
  kable(
    caption = "Maryland County Socioeconomic Indicators (2022)"
  ) |>
  kable_classic(
    lightable_options = "striped"
  ) |>
  kable_styling(
    latex_options = "scale_down",
    font_size = 9
  )

# ---- Author: Blake Naluai ----
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md_clean |>
  ggplot(
    mapping = aes(
      x = pct_internet,
      y = pct_wfh,
      size = pct_65_plus,
      color = median_income
    )
  )

```

```

    )
  ) +
  geom_point(alpha = 1) +
  facet_wrap(~year) +
  scale_x_continuous(labels = percent_format()) +
  scale_y_continuous(labels = percent_format()) +
  labs(
    caption = "U.S. Census Bureau ACS 5-year estimates via tidycensus",
    x = "% Internet Access",
    y = "% Working From Home",
    size = "% Age 65+",
    color = "Median Income"
  ) +
  theme_bw()

# ---- Author: Raj Ramnani ----
# ---- Reviewer: Blake Naluai & Raj Ramnani ----

housing_vars <- c(
  median_gross_rent = "B25064_001",
  renter_total = "B25070_001",
  rent_30_34 = "B25070_007",
  rent_35_39 = "B25070_008",
  rent_40_49 = "B25070_009",
  rent_50_plus = "B25070_010",
  total_housing_units = "B25002_001",
  vacant_housing_units = "B25002_003"
)

md_housing_2022 <- get_acs(
  geography = "county",
  state = "MD",
  variables = housing_vars,
  year = 2022,
  survey = "acs5",
  output = "wide"
) |>
  transmute(
    County = str_remove(NAME, "\\s*.*$"),
    median_gross_rent = median_gross_rentE,
    pct_renters_30_plus = (
      rent_30_34E + rent_35_39E + rent_40_49E + rent_50_plusE
    )
  )

```

```

    ) / renter_totalE,
    pct_renters_35_plus = (
      rent_35_39E + rent_40_49E + rent_50_plusE
    ) / renter_totalE,
    pct_renters_50_plus = rent_50_plusE / renter_totalE,
    vacancy_rate = vacant_housing_unitsE / total_housing_unitsE
  ) |>
  arrange(County)

md_housing_2022 |>
  mutate(
    median_gross_rent = dollar(median_gross_rent),
    pct_renters_30_plus = percent(pct_renters_30_plus, accuracy = 0.1),
    pct_renters_35_plus = percent(pct_renters_35_plus, accuracy = 0.1),
    pct_renters_50_plus = percent(pct_renters_50_plus, accuracy = 0.1),
    vacancy_rate = percent(vacancy_rate, accuracy = 0.1)
  ) |>
  select(
    "County" = County,
    "Median Rent" = median_gross_rent,
    "Renters 30%+" = pct_renters_30_plus,
    "Renters 35%+" = pct_renters_35_plus,
    "Renters 50%+" = pct_renters_50_plus,
    "Vacancy" = vacancy_rate
  ) |>
  kable(
    caption = "Maryland County Rental Housing Stress Indicators, 2022"
  ) |>
  kable_classic(
    lighttable_options = "striped"
  ) |>
  kable_styling(
    latex_options = "scale_down",
    font_size = 9
  )
# ---- Author: Raj Ramnani ----
# ---- Reviewer: Blake Naluai & Raj Ramnani ----

## Set variables
gini_vars <- c(
  gini_index = "B19083_001"
)

```

```

## Get Gini Index data for 2019
md_gini_2019 <- get_acs(
  geography = "county",
  state = "MD",
  variables = gini_vars,
  year = 2019,
  survey = "acs5",
  output = "wide"
) |>
  mutate(year = 2019)

## Get Gini Index data for 2022
md_gini_2022 <- get_acs(
  geography = "county",
  state = "MD",
  variables = gini_vars,
  year = 2022,
  survey = "acs5",
  output = "wide"
) |>
  mutate(year = 2022)

## Bind and tidy Gini data
md_gini_clean <- bind_rows(md_gini_2019, md_gini_2022) |>
  transmute(
    GEOID,
    County = NAME,
    year,
    gini_index = gini_indexE
  )

## Calculate county-level Gini Index change
md_gini_change <- md_gini_clean |>
  pivot_wider(
    names_from = year,
    values_from = gini_index,
    names_prefix = "gini_"
  ) |>
  mutate(
    gini_change = gini_2022 - gini_2019
  )

```

```

## Get Maryland county map boundaries
md_counties <- counties(state = "MD", cb = TRUE, year = 2022) |>
  st_as_sf()

## Join Gini change data to county map boundaries
md_gini_change_map <- md_counties |>
  left_join(md_gini_change, by = "GEOID")

## Create Gini Index change choropleth map
ggplot(md_gini_change_map) +
  geom_sf(aes(fill = gini_change), color = "gray50", linewidth = 0.2) +
  scale_fill_gradient2(
    low = "#4575B4",
    mid = "#F7F7F7",
    high = "#D73027",
    midpoint = 0,
    labels = label_number(accuracy = 0.001)
  ) +
  labs(
    subtitle = "Gini Index change, ACS 5-year estimates, 2019 to 2022",
    fill = "Gini Index\\nchange",
    caption = "Source: U.S. Census Bureau ACS 5-year estimates via tidycensus"
  ) +

  theme_minimal() +
  theme(
    axis.text = element_blank(),
    axis.title = element_blank(),
    panel.grid = element_blank(),
    legend.position = "left",
    plot.subtitle = element_text(hjust = 0.25),
  )

# Narrative:
# The Gini Index map shows that income inequality changed unevenly across Maryland
# counties from 2019 to 2022. Most counties show only very small changes, suggesting
# that income inequality was relatively stable overall. Taken together, the statewide
# pattern appears to show a slight net reduction in income inequality, but the change
# is not evenly distributed across counties.

# ---- Author: Raj Ramnani ----
# ---- Reviewer: Blake Naluai & Raj Ramnani ----

```

```

## Set variables
wfh_vars <- c(
  total_workers = "B08301_001",
  work_from_home = "B08301_021"
)

## Get work-from-home data for each year
years <- 2019:2022

md_wfh_all <- map_dfr(years, \(yr) {
  get_acs(
    geography = "county",
    state = "MD",
    variables = wfh_vars,
    year = yr,
    survey = "acs5",
    output = "wide"
  ) |>
  mutate(year = yr)
})

## Tidy work-from-home data
md_wfh_clean <- md_wfh_all |>
  transmute(
    GEOID,
    County = NAME,
    year,
    total_workers = total_workersE,
    work_from_home = work_from_homeE,
    pct_wfh = work_from_home / total_workers
  )

## Calculate average work-from-home percentage across Maryland counties
wfh_avg <- md_wfh_clean |>
  group_by(year) |>
  summarise(
    avg_pct_wfh = mean(pct_wfh, na.rm = TRUE),
    .groups = "drop"
  )

## Add 2018 and 2023 to the axis without plotting values for those years
wfh_avg_plot_data <- tibble(

```

```

    year = 2018:2023
) |>
  left_join(wfh_avg, by = "year")

## Create work-from-home line plot
wfh_plot <- wfh_avg_plot_data |>
  ggplot(aes(x = year, y = avg_pct_wfh)) +
  geom_line(linewidth = 1, na.rm = TRUE) +
  geom_point(size = 3, na.rm = TRUE) +
  scale_x_continuous(
    breaks = 2019:2022,
    limits = c(2019, 2022)
  ) +
  scale_y_continuous(
    labels = percent_format(accuracy = 1),
    limits = c(0, NA)
  ) +
  labs(
    x = "Year",
    y = "% Working from home",
    caption = "Source: U.S. Census Bureau ACS 5-year estimates via tidycensus"
  ) +
  theme_minimal() +
  theme(
    plot.subtitle = element_text(hjust = 0.5),
    panel.border = element_rect(color = "gray60", fill = NA, linewidth = 0.6)
  )

# Narrative:
# The work-from-home visualization shows a clear and consistent increase in the
# average share of Maryland county workers working from home from 2019 through 2022.
# This shows that remote work became much more common during this period,
# reflecting major world-wide changes consistent with the social distancing requirements.
wfh_plot

```


References:

- ASU EdPlus. (n.d.). In *ASU EdPlus Image Accessibility Creator*. Retrieved May 6, 2026, from https://asuo-ai-labs.streamlit.app/Image_Accessibility
- Walker, K. (n.d.). *Tidycensus*. Retrieved May 6, 2026, from <https://walker-data.com/tidycensus/index.html>